

Name: _____

Registration: _____

(10 pts.) Given the following partial module for a 512Kx64 RAM chip, fill in the blanks to complete the module.

```
module RAM (adr, CS, RW, di, do);

    input CS, RW;
    input [      ] adr;
    input [      ] di;
    output [      ] do;
    reg [      ] d_out;
    reg [      ] Mem1 [      ];

    assign do = (CS && RW)?d_out: _____;

    always @(adr or di or CS or RW)
        if (CS && !RW)
            Mem1 [      ] = _____;
    always @(adr or CS or RW)
        if (CS && RW)
            d_out = Mem1 [      ];

    initial
        $readmemh ("memory1.dat", Mem1);

endmodule
```

Name: _____

Registration: _____

(10 pts.) Given the following partial module for a 256Kx128 RAM chip, fill in the blanks to complete the module.

```
module RAM (adr, CS, RW, di, do);

    input CS, RW;
    input [      ] adr;
    input [      ] di;
    output [      ] do;
    reg [      ] d_out;
    reg [      ] Mem1 [      ];

    assign do = (CS && RW)?d_out: _____;

    always @(adr or di or CS or RW)
        if (CS && !RW)
            Mem1 [      ] = _____;
    always @(adr or CS or RW)
        if (CS && RW)
            d_out = Mem1 [      ];

    initial
        $readmemh ("memory1.dat", Mem1);

endmodule
```